Biodiversity data: From data collection to publication

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based on FRB-CESAB / PNDB / GBIF France training course course

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https://biodiversitydata.github.io/

Goal: Work on the different stages of the data lifecycle, from acquisition to opening, including management, storage and the drafting of data papers.

We will

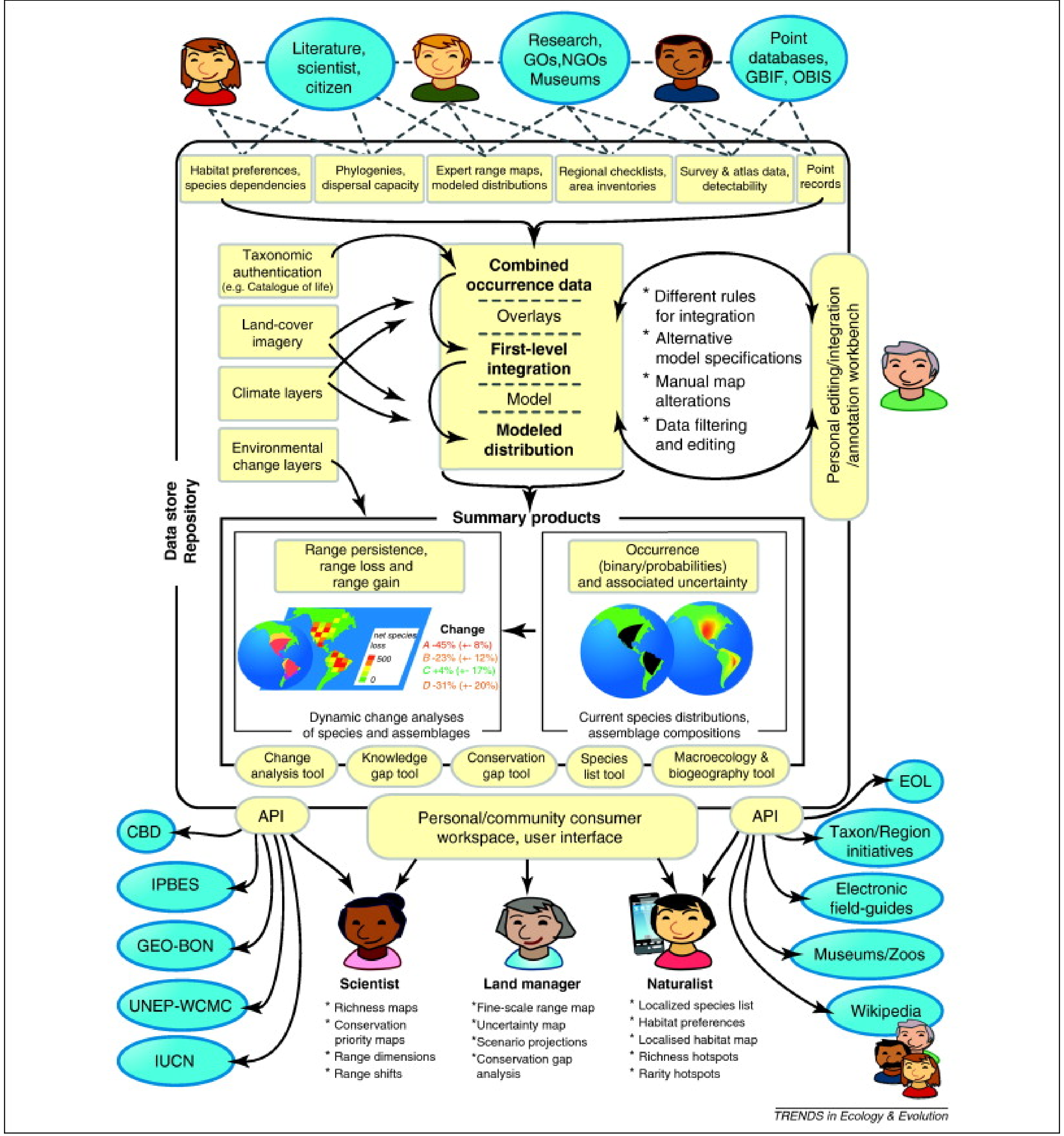
1) contextualize the issues surrounding the understanding, sharing and (re)use of biodiversity data and metadata,
2) enhance the skills of communities involved in one or more stages of the data lifecycle.

Opinion

Integrating biodiversity distribution knowledge: toward a global map of life

Walter Jetz[1], Jana M. McPherson[2,3] and Robert P. Guralnick[4,5]

Schematic diagram showing how producers and users of biodiversity information interact.



Program

Day 1:

General context

- Current challenges and data ecosystem
- Reproducibility concepts
- Major types of biodiversity data
- Framework and good practices, Data management plan

Data Acquisition Part 1:

- Best practices for data collection
- Focus GBIF (introduction to Standardization, Data origin and Types, Quality)
- Exercise

Day 2:

Data Acquisition Part 2:

- International repositories
- Major biodiversity databases available

Practical:

- Web portals, API & web scraping
- Eternal problem of taxonomy

Day 3:

Data Management:

- Structure, formats & files
- Cleaning data CheckList

Practical:

- Introduction to Tidyverse (dplyr, stringr, tidyr, ggplot2)

Day 4:

Geographic information:

- Spatial data
- Major environmental databases available

Practical:

- R spatial introduction

Day 5:

Opening data:

- Metadata and standards, Storage and archiving
- Dissemination and sharing, Data & software papers
- Overview of Reproductible research tools (Rmarkdown, Docker, Git, Rpackages)